# Base station physics: summary

Params: base/spec/params.py v1.2.8

### Spring (each, A228, SF 1.2 on Shigley static limits)

| | |
|---|---|
| Rate used / by formula | 0.125 / 0.117 N/mm |
| Initial tension used | 7.0 N |
| Body coils, mass | 98, 53 g |
| Safe force (governed by) | 40.9 N (hook torsion) |
| Safe extension (wire) / spec cap | 271 / 265 mm |
| Surge (fixed-fixed) | 24 Hz |

### Launch (chain spring model)

| | |
|---|---|
| Stroke, ratio, pitch | 241 mm, 1.0:1, 35 deg |
| Spring extension full / rest | 265 / 24 mm |
| Stored energy (2 springs) | 12.1 J |
| Carriage force at full draw | 80 N |
| Muzzle speed | 6.94 m/s |
| Peak acceleration, stroke time | 19 g, 58 ms |
| Efficiency (dart KE / stored) | 56 % |
| Carriage KE at stop, pad peak | 1.69 J, 265 N (19 mm pad) |
| Spring KE at stop, coil clash | 0.83 J, 8.7 m/s |

### Flight (bead-chain tether + reel model, level ground)

| | |
|---|---|
| Line path at the shot | taut (path only lengthens) |
| Range at 0.5 / 0.75 / 1.0 N brake | 4.28 / 3.85 / 3.47 m (pitch 34) |
| Same shot, no tether | 5.13 m |
| Flight time, apex (0.75 N) | 0.81 s, 1.12 m |
| Impact speed, angle (0.75 N) | 5.8 m/s, 55 deg down |
| Ballistics table | 6 to 47 deg, best 33.5 deg: 3.85 m |

### Storage reel payout (LIGHT 0.5 N at full reel)

| | |
|---|---|
| Reel empty mass, inertia full | 46 g, 31.4 kg mm2 |
| Line radius full / core | 30.7 / 25.0 mm (brake +23 % at core) |
| Line pull in the stroke (spin-up) | 5.18 N |
| Reel rpm max | 2158 rpm |
| Overrun after landing | 0.58 m in 0.27 s |

### Capstan and tow

| | |
|---|---|
| Wraps for tow limit (mu 0.15) | 2.4 raw, 4 design |
| Take-up needed (design wraps) | 0.07 N |
| Tow at limit: torque, speed, current | 4.17 N m, 0.128 m/s, 6.09 A |
| Line pull at limiter (5 N m) (12.6 V) | 61 N (break 667 N) |
| Line pull, motor stall (no limiter, 12.6 V) | 174 N |
| Tow sensing: line N per wiper amp at 11.3 V (line = 11.3 x (I - I_free) - 2.0 N) | |

### Drivetrain (wiper at 11.1 V with sag)

| | |
|---|---|
| aim: peak torque, current, time | 0.33 N m, 1.9 A, 7 s |
| cock: peak torque, current, time | 2.93 N m, 4.7 A, 3 s |
| reel: peak torque, current, time | 0.43 N m, 2.0 A, 107 s |
| tow: peak torque, current, time | 4.17 N m, 6.1 A, 39 s |
| Wiper at limiter (5 N m), 12.6 V (with sag) | 5.0 N m, 7.0 A |
| Wiper motor stall (no limiter), 12.6 V (with sag) | 14.1 N m, 17.0 A |

### Harry must measure (then edit the parameter block and rerun: python summary.py)

- Spring hang test: hang 1, 2, 3 kg from one spring, measure the length each time. Slope = rate, intercept = initial tension. Every range number scales with the stored energy.
- Wiper: free speed (turns in 30 s at 12.0 V), free current, stall torque (arm + luggage scale, 1 s) and stall current. Edit common.Wiper or spec WIPER.
- Dart mass on a kitchen scale with ballast and tether knot (spec 0.279 kg).
- Slip clutch: luggage scale on the line with the reel full: LIGHT 0.5 N, FIRM 5 N.
- After the first shots: python calib.py range pitch:brake:range ... then python ballistics.py --v-scale <factor>.
- Rail friction: tilt the rail with the dart on it until it slides (mu = tan angle).